Novus ADOS — Technical Architecture & Executive System Specification

Autonomous Defect & Orchestration System *Enterprise Grade | Governed Agentic Swarm | Built on IBM watsonx*

Executive Summary

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  ■                                                                          ■
  ■   HUMAN COORDINATION TAX          NOVUS ADOS COMPRESSED LOOP            ■
  ■   ■■ 4.2 Hours Mean Time          ■ 39 Minutes Compressed MTTR          ■
  ■      To Resolution (MTTR)            (-84% Reduction in Line Stoppage)  ■
  ■                                                                          ■
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Novus ADOS (Autonomous Defect and Orchestration System) is an enterprise-grade AI decision stack engineered to solve the single largest operational inefficiency in discrete manufacturing: the **Human Coordination Tax**.

When a physical defect occurs on a high-precision production line—such as a bore-tolerance excursion on an Electric Vehicle (EV) motor housing—the physical fault triggers hours of manual cross-departmental coordination across engineers, managers, buyers, and plant leads. Novus ADOS replaces this manual triage loop with an autonomous, 6-layered agentic decision stack that detects, diagnoses, simulates, governs, and executes incident recovery in **minutes instead of hours**.

The Governed Autonomy Paradigm: Unlike traditional "AI recommendation tools" that stop short of system action, Novus ADOS operates within strict financial exposure and risk boundaries. High-confidence, low-cost actions execute autonomously at **Tier 0**, while high-cost or lower-confidence actions route seamlessly to human decision-makers via **Tier 1/2 Governance Locks**—with every single action logged to an immutable audit trail.

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1. The 6-Layer Enterprise Decision Stack

The Novus ADOS architecture is organized into **6 modular layers**, providing complete decoupling between low-level telemetry ingestion and C-suite operational intelligence.

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graph TD
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    classDef l5 fill:#4c1d95,stroke:#a855f7,stroke-width:2px,color:#fff;
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    L6["L6 :: Executive Intel (watsonx.ai Copilot, MTTR Analytics, Revenue-at-Risk)"]:::l6
    L5["L5 :: Governance Engine (Tier 0 Autonomy, Tier 1/2 Approval Locks)"]:::l5
    L4["L4 :: Decision Orchestrator (State Machine, ServiceNow, SAP RFC)"]:::l4
    L3["L3 :: Decision Memory (Cloudant Vector Precedent RAG, Bayesian Graph)"]:::l3
    L2["L2 :: Specialist Agent Swarm (Vision, CAD, Causal, Param, Sub, Sim)"]:::l2
    L1["L1 :: Event Bus & Envelope (FastAPI Ingress, Redis Pub/Sub Payload)"]:::l1

    L6 <--> L5
    L5 <--> L4
    L4 <--> L3
    L3 <--> L2
    L2 <--> L1
```

Layer Breakdown Matrix

Layer	Module Name	Primary Responsibilities	Core Tech Stack
L6	**Executive Intel**	Plant health index, revenue-at-risk calculation, watsonx.ai Copilot natural language interface.	`watsonx.ai`, Next.js 15, Tailwind
L5	**Governance Engine**	Risk classification ($ Exposure × Confidence), RBAC session validation, Tier 0/1/2 policy locks.	`orchestrate/governance.py`, JWT
L4	**Decision Orchestrator**	Multistage state machine, live ServiceNow ticket creation, SAP ERP inventory reservations.	`watsonx Orchestrate ADK`, Python
L3	**Decision Memory**	RAG precedent vector indexing, Bayesian causal edge weight recalibration.	`IBM Cloudant NoSQL`, Cosine RAG
L2	**Specialist Agent Swarm**	8 L2 specialist agents for perception, CAD alignment, causal isolation, and Monte Carlo simulation.	`Local LLM (Qwen3)`, OpenCV
L1	**Event Bus & Envelope**	Fast low-latency event serialization, micrometer anomaly payload schema validation.	`FastAPI`, Pydantic v2, Redis

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2. End-to-End Scenario: Nova Motors EV Plant

Case Study Context

- **Facility**: Nova Motors EV Manufacturing Plant #04 (Line B)
- **Target Part**: Aluminum Motor Housing (Part #`MH-409`)

- **Incident Trigger**: CNC Spindle #04 Bore-Tolerance Breach (`+0.42 mm` deviation vs < `0.15 mm` tolerance limit).

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  TRADITIONAL MANUAL LOOP (4.2 HOURS)
  [Anomalous Bore] ■■> [Manual Metrology Inspection] ■■> [Engineering Huddle] ■■> [Manual SAP
Check] ■■> [ServiceNow Ticket]

  NOVUS ADOS COMPRESSED PIPELINE (39 MINUTES)
  [FastAPI Ingress] ■■> [L2 Swarm Causal LLM] ■■> [Monte Carlo Sim] ■■> [L5 Governance Gate]
■■> [Live watsonx ServiceNow Action]
```

Phase-by-Phase Technical Pipeline

Phase I: Anomaly Detection & Causal Isolation

- **L1 Event Ingress**: Sensors on CNC Spindle #04 register vibration harmonics. FastAPI ingests the JSON payload (`evt-77a8b-spindle-4`) and publishes it to the event bus.
- **L2 Perception & CAD Spec**: *Vision Spec Agent* scans the bore surface while *CAD Spec Agent* calculates micrometer offset vectors against nominal STEP CAD files (`offset = +0.421 mm`).
- **L2 Causal Isolation & L3 Memory**: The *Causal Isolation Agent* traverses the Bayesian causal graph, querying **IBM Cloudant** precedent vectors. Combining graph probabilities with local LLM reasoning over shift logs, it isolates the primary root cause: **Spindle Bearing Thermal Expansion (87% probability)**.

Evidence Chain Evaluated: Spindle Vibration (`0.42mm`), Tool Wear Count (`1,420 cycles`), Material Batch (`#BATT-992`), Ambient Humidity (`62%`).

Phase II: Simulation & Option Ranking

- **Option Generation**:
- **Option A (Parameter Adjustment)**: Reduce spindle speed by 15% and increase coolant flow by 2.0 bar (Zero part replacement cost; **Estimated Downtime: 12 mins**).
- **Option B (Part Substitution)**: Swap spindle cartridge assembly `#SP-04-B` via SAP inventory (**Cost: $1,250; Estimated Downtime: 45 mins**).
- **L2 Impact Simulation**: Runs Monte Carlo financial simulations across candidate options to project MTTR, scrap rate, component cost, and quality risk scores.

Phase III: Governance & Real Enterprise Execution

- **L5 Governance Tier Determination**: The system evaluates policy rules:

$$\text{Financial Risk} = \text{Option Cost} \times (1 - \text{Confidence Score})$$

- *Option A* ($0 exposure, 92% confidence) $\rightarrow$ **Tier 0 Fully Autonomous**.
- *Option B* ($1,250 cost, Tier 1 policy boundary) $\rightarrow$ **Routes to Plant Manager Approval Queue**.
- **L4 System Execution (watsonx Orchestrate)**: Upon human operator approval of Option B:
- Dedicated `ados_itsm_agent` calls live **IBM watsonx Orchestrate ADK** tools.
- Executes authenticated Table API writes to **ServiceNow** (`INC0094821` & `CHG0048201`).

• Issues SAP RFC inventory reservation for replacement cartridge `#SP-04-B`.

Phase IV: Bayesian Calibration Learning

• **L3 Feedback Calibration**: Once resolved, the *Feedback & Calibration Agent* logs the verdict to Cloudant, updating Bayesian edge weights so future root-cause confidence improves automatically.

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3. IBM Technology Stack Integration

Novus ADOS is natively integrated into the **IBM Ecosystem (Build on watsonx)**:

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 ■                         IBM WATSONX ECOSYSTEM                         ■
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 ■   ■  watsonx Orchestrate    ■         ■    IBM Cloudant NoSQL DB    ■   ■
 ■   ■  - ADK Tool Bindings    ■         ■    - Precedent Vectors        ■   ■
 ■   ■  - ServiceNow Table API■         ■    - JSON Incident Logs       ■   ■
 ■   ■■■■■■■■■■■■■■■■■■■■■■■■■■■         ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■   ■
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 ■   ■    IBM Watson NLU       ■         ■    IBM Watson Speech (TTS)    ■   ■
 ■   ■  - Log Entity Mining    ■         ■    - Allison Voice Synthesis ■   ■
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Live ServiceNow Write Authentication: Unlike simulated demos, Novus ADOS uses production IAM bearer tokens (`WO_API_KEY`) and live watsonx Orchestrate endpoints (`api.br-sao.watson-orchestrate.cloud.ibm.com`) to execute real system-of-record writes.

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4. Multi-Role RBAC & User Experience

Novus ADOS enforces strict **Role-Based Access Control (RBAC)** across four distinct enterprise personas:

| Persona | Primary Interface | Permissions & Capabilities | Key UX Goal |

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| **Operator** | Line Control Panel | View real-time line alarms, execute Tier 0 overrides, receive spoken TTS incident briefings. | Zero friction on plant floor |

| **Engineer** | Swarm Graph & Studio | Inspect L2 agent contracts, analyze Bayesian causal graph edges, trigger manual diagnostic runs. | Deep root-cause explainability |

| **Executive** | Executive Dashboard | Monitor revenue-at-risk, plant availability index, MTTR compression metrics, chat with watsonx Copilot. | High-level strategic oversight |

| **Auditor** | Governance Log | Inspect cryptographic decision lineage, verify approval sign-offs, audit policy compliance. | Complete regulatory traceability |

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5. Quantifiable ROI & Business Impact

Based on Nova Motors EV Plant operational data:

| Metric | Traditional Manual Process | Novus ADOS Autonomous Loop | Net Improvement |

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| **Mean Time to Resolution (MTTR)** | 4.2 Hours (252 mins) | **39 Minutes** | **■ -84% Reduction** |

| **Human Coordination Tax** | $3,850 / incident | **$210 / incident** | **■ 94% Cost Reduction** |

| **Monthly Revenue Protected** | Baseline | **+$142,500 / month** | **■ Direct Top-Line Recovery** |

| **Autonomous Resolution Index** | 0% | **94% Tier 0 Autonomy** | **■ Minimal Shift Interruption** |

Production Safety Lock: Live write capabilities to external ERP and ITSM systems are governed by the `WO_ITSM_LIVE_WRITES_ENABLED=true` feature flag. Always verify test environment credentials before enabling live SAP write execution.